EXECUTIVE SUMMARY

TURNING SURVIVAL HOURS INTO DAYS.

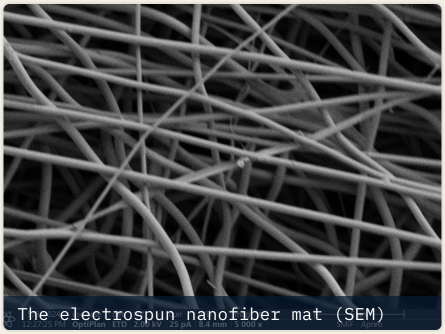
When evacuation is hours or days away, survival depends on pain control that lasts that long without sedating the casualty. Opioids dull pain but do exactly that; even long-acting local anesthetics fade within about a day.

> **“What was the golden hour? I argue all the time […] in a peer-to-peer conflict, especially in the Pacific […] it may be a golden day or golden week.”**
>
> **Brig. Gen. James R. Parry** · Chief of Staff, Michigan National Guard; former USAF Director of Medical Operations

> **“I had to move him overland by about six hours of driving. I had to repeatedly stop and dose for pain […] Being able to just manage pain without having to redose […] would have been a huge mental load off of my plate.”**
>
> **Former SOF 18-Delta paramedic** · austere & remote trauma care

X49 is a localized, opioid-free nanofiber designed to be placed in the field, even by a non-medic, and to control pain at the wound for days, keeping the wounded conscious, mobile, and survivable until evacuation.

WHY IT KEEPS WORKING IN AN ACIDIC WOUND.


The electrospun nanofiber mat (SEM)

1 IT OUTLASTS THE ACID
An injured wound bed turns acidic, which blunts standard local anesthetics. Even the long-acting options give out within a day. A sodium-bicarbonate additive is engineered to sustain release in that environment, so pain control holds where it matters.

2 LOCALIZED, TUNABLE, RESORBABLE
A drug-loaded matrix releases the painkillers at the wound, tunable for duration, then dissolves over weeks to months. No opioids, no needles, no second surgery to remove it.

1	DAYS	0
APPLICATION	OF PAIN CONTROL	OPIOIDS
Packed into the wound by hand, in the field.	One dose holds at the wound. No redosing under fire.	No overdose risk; the casualty stays in the fight.

WHAT THE EARLY STUDY SHOWS.

~40%
PAIN REDUCTION

In an early **live tissue study**, the nanofiber reduced pain to about 60 percent of normal while preserving motor function. We are not erasing pain. **We are taking the edge off and keeping the casualty functional until evacuation.**

	TODAY opioids / current local anesthetics	**WITH X49** intended capability, pre-human trial
Casualty stays conscious and able to move	No	Yes
Works inside an acidic wound	No	Yes
No needles; placeable without a medic	No	Yes

> **“Future conflicts may not allow the rapid evacuation timelines we’ve become accustomed to.”**

Col Nicole Christiano · Commander, 146th Medical Group

WHERE WE ARE

- The polymer already has an FDA Drug Master File on the registry (a regulatory filing, not approval).
- Active drugs are decades-old generics; meloxicam is in the combat wound medication pack.
- Funded by the U.S. Department of Defense (ASD for Health Affairs) through the CDMRP Peer Reviewed Medical Research Program.

WHAT’S UNDERWAY

- **JUL 2026** Live tissue study; results mid-July.
- **AUG 2026** Northern Strike participation; brief the Air Force Surgeon General.
- **SEP 2026** NIH SBIR submission. FDA submission target ~2028.

THE ASK

We are seeking **$1M to $1.5M in bridge funding before October 2026** to complete the preclinical efficacy and safety package and pull FDA human-trial readiness forward 6 to 9 months, toward a 2028 submission. Path: SBIR/STTR, then cross-service TACFI/STRATFI.

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